Derivation of a mouth state index using Bernoulli's equation

Intantaneous analysis

Start with Bernoulli's equation for incompressible flow to derive an equation for the flow of water from an estuary through its inlet to the ocean.

$$P + \frac{1}{2}\rho v^2 + \rho g h = \text{constant} \tag{1}$$

where P is the pressure, ρ is the fluid density, v is the fluid velocity, g is the acceleration due to gravity, h is the height above a datum.

Consider two points: point 1 at the estuary's surface with height h_{est} above the inlet, and point 2 at the inlet. Assuming:

- The flow is steady.
- The velocity at the estuary surface (point 1) is much less than at the inlet (point 2), so $v_1 \approx 0$.
- The pressure at both points is atmospheric, so $P_1 = P_2$.
- The inlet is at sea level, so $h_{inlet} = 0$.

Applying Bernoulli's equation between the surface of the estuary and the inlet to the ocean, we get:

$$P_{est} + \frac{1}{2}\rho v_{est}^2 + \rho g h_{est} = P_{inlet} + \frac{1}{2}\rho v_{inlet}^2 \tag{2}$$

The equation simplifies to:

$$\rho g h_{est} = \frac{1}{2}\rho v_{inlet}^2 \tag{3}$$

Solving for the velocity at the inlet:

$$v_{inlet} = \sqrt{2 g h_{est}} \tag{4}$$

To account for inlet resistance, introduce a drag coefficient (C_d) to account for energy losses and flow contraction:

$$v_{inlet} = C_d \sqrt{2 g h_{est}} \tag{5}$$

C_d takes into account factors such as the shape of the estuary mouth and the roughness of the boundaries.

The flow rate (Q) through the inlet can be calculated as:

$$Q = A_{inlet} \cdot v_{inlet} = C_d A_{inlet} \sqrt{2 g h_{est}} \tag{6}$$

where A_{inlet} is the cross-sectional area of the estuary inlet. Our goal is to determine the effective resistance of an estuary mouth and predict the time it will take for a flooded estuary to drain into the ocean.

We could fit Equation 6 directly to the data to estimate the instantaneous $C_d A_{inlet}$. An event-scale analysis would give us an integrated $C_d A_{inlet}$ that considers a high flow/flood event as a whole.

Event scale analysis

Determine the time it would take to drain the estuary by considering the volume of water in the estuary and the rate of flow out to the sea, and any additional water from river inflow:

$$t = \frac{V}{Q - Q_{inflow}} \tag{7}$$

where t is the time to drain, V is the volume of water to be drained, Q is the flow rate out of the estuary, Q_{inflow} is the inflow rate of the river (or other sources) into the estuary.

Combining with Equation 6,

$$t(C_d A_{inlet}\sqrt{2gh_{est}} - Q_{inflow}) = V \tag{8}$$

$$C_d A_{inlet} = \frac{V/t + Q_{inflow}}{\sqrt{2gh_{est}}} \tag{9}$$